

SMART BATTERY REUSE IDENTIFICATION SYSTEM

Battery Condition Analysis Report

Report ID: RPT-20260509112741 · 09 May 2026, 11:27:41

1. BATTERY DETAILS

Parameter	Value
Battery ID	BAT-001
Measured Voltage	3.70 V
Analysis Date	09 May 2026
Analysis Time	11:27:41
Report ID	RPT-20260509112741

2. IMAGE PREPROCESSING STEPS

#	Processing Step	Status
1	Grayscale conversion	SIMULATED
2	Gaussian blur (kernel 5×5)	SIMULATED
3	ROI extraction (80% crop)	SIMULATED
4	Edge detection (Canny)	SIMULATED

3. DEFECT ANALYSIS

Defect Type	Status	Severity	Confidence
Swelling	CLEAR	—	—
Corrosion	CLEAR	—	—
Burn Marks	CLEAR	—	—
Label Damage	DETECTED	Low	63%

4. CLASSIFICATION RESULT

Classification Grade	Grade B — Usable with Caution
Recommendation	Minor defect detected (Label Damage). Battery is usable with caution. Monitor temperature during use.

This report was generated by the Smart Battery Reuse Identification System. Results are based on image analysis and voltage measurements. Final reuse decisions must be confirmed by a qualified engineer.